

Module 1: Biology - The Study of Life - Practice Quiz

Use this low-stakes quiz after reviewing the module keys and learning questions.

1. A student sees frost crystals slowly "grow" on a cold window overnight. Why don't biologists count this as life?

- A. Growth by itself is not enough; life needs cells, metabolism, response, and reproduction together.
- B. Crystals are too small to be considered alive.
- C. Crystals are never found in the natural world.
- D. Only warm objects can ever be alive.
- Answer: A
- Why: No single trait defines life. Living things share a set of properties together: cells, metabolism, homeostasis, response, reproduction, and evolution.

2. To test whether fertilizer speeds plant growth, a student changes the amount of fertilizer given to each plant. The amount of fertilizer is the:

- A. hypothesis being tested
- B. variable that is changed
- C. final conclusion
- D. supporting theory
- Answer: B
- Why: The variable is the factor that changes in an investigation. The hypothesis is the testable explanation, and the conclusion comes after the data.

3. Which list places levels of biological organization from smallest to largest?

- A. Ecosystem, organism, cell
- B. Organism, cell, molecule
- C. Cell, organism, ecosystem
- D. Population, cell, molecule
- Answer: C

- Why: Biology is organized across scales: cells build organisms, which form populations and ecosystems.

4. A hypothesis survives many careful tests over many years. What should happen to it?

- A. It becomes a fixed fact that can never change.
- B. It should be hidden so no one can disprove it.
- C. It must be discarded simply because it is old.
- D. It may contribute evidence to a broader theory, but hypotheses and theories are different kinds of scientific explanation.
- Answer: D
- Why: A hypothesis is a testable explanation for a specific observation or pattern. A theory is a broad explanation supported by many lines of evidence; a hypothesis does not simply mature into a theory. Both remain open to revision when new evidence appears.